THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is the best explanation for the cooling in a desert cooler? 1

(a) fan makes cold
(b) pads produce cold
(c) water is cold
(d) water evaporates and takes heat from the air

2. The best way to dry a wet handkerchief quickly is to: 1

(a) spread it open in warm, dry, windy place **(b)** dip it in cold water
(c) fold it and keep in a closed box **(d)** keep in a damp room

3. Assertion (A): Naphthalene balls disappear without leaving liquid. Reason (R): They undergo sublimation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A beaker of water is heated and bubbles appear at the bottom well before 100 degrees Celsius. They are mainly: 1

(a) steam **(b)** dissolved air coming out
(c) oxygen from water **(d)** hydrogen

5. Camphor kept in an open dish disappears but wax melts. This is because camphor: 1

(a) sublimes, wax melts then remains **(b)** freezes
(c) melts like wax **(d)** burns

6. Assertion (A): Water boils at a lower temperature on a mountain top. Reason (R): Atmospheric pressure is lower at the top. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Naphthalene and ammonium chloride can both be separated from sand by: 1

(a) filtration **(b)** evaporation **(c)** decantation **(d)** sublimation

8. A gas is cooled until it turns into a liquid and then a solid. Compared with the gas, the solid has: 1

(a) no forces **(b)** particles far apart
(c) faster particles **(d)** particles close and in fixed positions

9. **Assertion (A):** A solid has a definite shape. 1
Reason (R): The particles of a solid can move freely past one another.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The particles in a solid do not move at all. 1
Reason (R): The particles in a solid are held together by strong forces.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How can you separate a mixture of common salt and camphor? Which property is used? 2

12. How can the boiling point of water be raised and how can it be lowered? 2

13. Why does the temperature of boiling water not rise even when the flame is increased? 2

14. Explain why the particles in a liquid do not fly apart like those in a gas. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A glass of cold water on a table has water droplets on its outer surface. Explain the origin and why it appears more on a humid day. 3

16. Explain the process of how perfumes work using diffusion and evaporation. 3

17. Explain why it is easier to compress a gas than a liquid with the help of the data: air can be compressed to 1/10th, water practically not. 3

OR

Explain why spirit evaporates faster than water and cools more. 3

18. A mixture of two liquids is heated and one vaporises first. Name the concept and its use. 3

OR

Why does a camphor tablet get smaller on a hot day faster than on a cold day? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A student heats crushed ice from -10 degrees C steadily and records the temperature. It rises to 0 degrees C, stays at 0 degrees C until all the ice melts, then rises again to 100 degrees C, stays at 100 degrees C while the water boils, and rises further as steam is heated. 5
(i) What is the melting point of ice?
(ii) What is the boiling point of water at normal pressure?
(iii) What is the temperature of the mixture while ice is melting?
(iv) The temperature does not rise during melting even though heat is supplied. Explain.

20. Discuss the role of evaporation in the water cycle and in regulating body temperature. Explain with particles. 5

OR

Describe how the properties of solid, liquid and gas decide their uses: giving two examples each. 5

END OF QUESTION PAPER — Total: 40 Marks